

Time Series Analysis

Chapter 2: ARMA models



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **EU**: European Union
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **INS**: Institutul Național de Statistică — the National Institute of Statistics (Romania)
- **LB**: Ljung–Box (test)
- **MA**: Moving Average
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **OPG**: Outer Product of Gradients
- **PACF**: Partial AutoCorrelation Function
- **QQ**: Quantile–Quantile (plot)
- **VAR**: Vector AutoRegression

Today's question and route

- **Question**: a stationary series is correlated with its own past; which small model captures this memory and turns it into forecasts with honest uncertainty?
  - the answer of Box and Jenkins: **ARMA(p, q)** models, a few parameters instead of infinitely many Wold weights
- **Route** of the chapter
  - AR(p) and MA(q): stationarity, invertibility, characteristic roots, ACF and PACF
  - ARMA(p, q): the causal MA(∞) form, the impulse responses, identification
  - estimation (Yule–Walker, conditional least squares, maximum likelihood), AIC and BIC, residual diagnostics
  - forecasting and the Box–Jenkins method on Romanian GDP, inflation, BET, EUR/RON and the sunspots
- Seminar 2 comes before this lecture: it gives the formulas; here we derive them and use them on data

Learning outcomes

- Check the stationarity of an AR(p) and the invertibility of an MA(q) from the roots of their polynomials
- Derive the mean, the autocovariances and the ψ weights of AR(1), AR(2), MA(q) and ARMA(1,1)
- Propose candidate orders from the ACF and the PACF, and choose among them with AIC and BIC
- Estimate ARMA models by Yule–Walker, conditional least squares and maximum likelihood, and read the output
- Test the residuals: Ljung–Box with $m - p - q$ degrees of freedom, Jarque–Bera
- Compute ARMA forecasts and their intervals, and explain mean reversion

Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 3–4 (stationary models, ARMA)
  - companion, free online: Hyndman and Athanasopoulos (2021), Ch. 9 (ARIMA models)
- Theory: Brockwell and Davis (2016), Ch. 3 (ARMA) and Ch. 5 (estimation); Hamilton (1994), Ch. 3–5; the classic: Box, Jenkins and Reinsel (2008)
- Python Quantlets of this chapter: Quantlets/Ch_02
  - ARMA estimation with `statsmodels` (`ARIMA`, `ArmaProcess`, `acorr_ljungbox`)
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Tools from Chapter 1

- **Weak stationarity**: constant mean μ and autocovariance $\gamma(h) = \mathrm{Cov}(X_t, X_{t+h})$ that depends only on the lag h
  - ACF $\rho(h) = \gamma(h)/\gamma(0)$; PACF ϕ_{hh} = correlation of X_t and X_{t+h} after removing $X_{t+1}, \ldots, X_{t+h-1}$
- **White noise** $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$: mean 0, variance σ^2, no autocorrelation
  - the **lag operator**: $LX_t = X_{t-1}$, $L^k X_t = X_{t-k}$; polynomials in L act on series
- **Wold decomposition** (Wold (1938)): every stationary, purely nondeterministic process is $X_t = \mu + \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$, $\psi_0 = 1$, $\sum_j \psi_j^2 < \infty$
  - details and charts: Chapter 1, sections on the lag operator, Wold and the sample ACF and PACF

The problem: infinitely many weights

- Wold says **what** a stationary process looks like, not **how to estimate it**
  - $\psi_1, \psi_2, \ldots$: infinitely many unknowns, and only T observations
- **Idea**: write $\psi(L) = \sum_j \psi_j L^j$ as a **ratio of two short polynomials**
  - $\psi(L) = \theta(L)/\phi(L)$ with $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$, $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$
  - then $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$: the **ARMA$(p, q)$** model with $p + q + 2$ parameters (ϕ, θ, μ, σ^2)
- A rational function approximates almost any smooth $\psi(L)$ well with small p and q
  - the principle of **parsimony**: the smallest model that leaves white-noise residuals

Three building blocks

- **AR(p)**, autoregressive: today depends on the last p values of the series
  - $X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$; origin: Yule (1927), sunspots
- **MA(q)**, moving average: today depends on the last q shocks
  - $X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$; origin: Slutzky (1937), sums of random causes
- **ARMA(p, q)**: both parts; popularised as a complete method by Box, Jenkins and Reinsel (2008) (first edition 1970)
  - c is the **intercept**; the mean is $\mu = c/(1 - \phi_1 - \cdots - \phi_p)$, not c
- Throughout: $\varepsilon_t \sim \text{WN}(0, \sigma^2)$, and for likelihood and intervals ε_t i.i.d. $N(0, \sigma^2)$ (independent and identically distributed)

The AR(1) model

- $X_t = c + \phi X_{t-1} + \varepsilon_t$, or $(1 - \phi L)X_t = c + \varepsilon_t$
  - ϕ: the share of today's deviation from the mean that survives until tomorrow
- **Back substitution** k times:
  - $X_t = c(1 + \phi + \cdots + \phi^{k-1}) + \phi^k X_{t-k} + \sum_{j=0}^{k-1} \phi^j \varepsilon_{t-j}$
  - if $|\phi| < 1$: $\phi^k \to 0$ and $X_t = \dfrac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$
- This is the Wold form with $\psi_j = \phi^j$: the **causal** (MA(∞)) representation
  - causal = X_t depends only on present and past shocks
  - $\phi = 1$: random walk (Chapter 1), no stationary solution; $|\phi| > 1$: explosive, or a solution that depends on future shocks, excluded

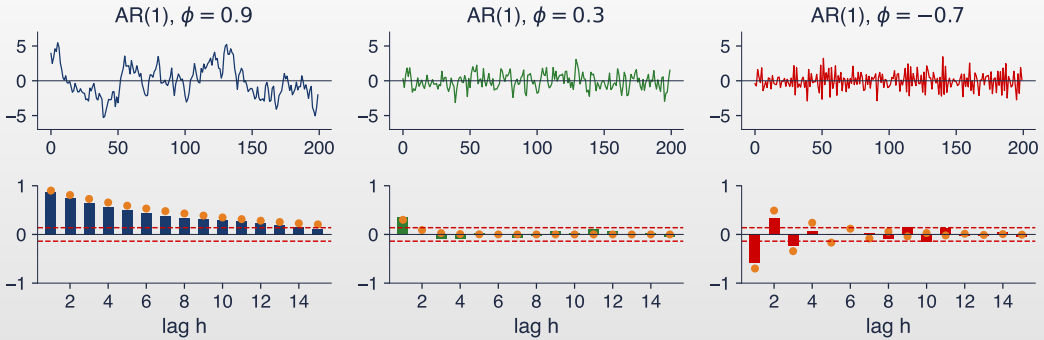
Moments of a stationary AR(1)

- **Mean**: take expectations, $\mu = c + \phi\mu$, so $\mu = c/(1-\phi)$
  - stationarity is used here: $E[X_t] = E[X_{t-1}]$
- **Variance**: $\gamma(0) = \phi^2\gamma(0) + \sigma^2$, since ε_t is uncorrelated with X_{t-1}
  - $\gamma(0) = \sigma^2/(1-\phi^2)$: grows without bound as $|\phi| \to 1$
- **Autocovariances**: multiply $X_t - \mu = \phi(X_{t-1} - \mu) + \varepsilon_t$ by $X_{t-h} - \mu$, $h \geq 1$, and take expectations
  - $\gamma(h) = \phi\gamma(h-1)$, so $\gamma(h) = \phi^h\gamma(0)$ and $\rho(h) = \phi^h$
  - geometric decay; alternating signs if $\phi < 0$
- **Half-life** of a shock: the h with $\phi^h = 1/2$, i.e. $h = \ln 0.5 / \ln \phi$

Worked example: an AR(1)

- $X_t = 2 + 0.8X_{t-1} + \varepsilon_t$, $\sigma^2 = 1$
- Stationary? $|\phi| = 0.8 < 1$: yes
  - mean $\mu = 2/(1 - 0.8) = 10$; the intercept is 2, the mean is 10
- Variance and ACF
  - $\gamma(0) = 1/(1 - 0.64) = 2.78$
  - $\rho(3) = 0.8^3 = 0.512$
- Half-life $= \ln 0.5 / \ln 0.8 = 3.11$ periods
  - after about 3 periods, half of a shock is gone
- ψ weights: $1, 0.8, 0.64, 0.512, \ldots$: a shock of 1 today moves X_{t+2} by 0.64

AR(1) for three values of ϕ



- The same 200 shocks, $\phi = 0.9$, 0.3, -0.7; bottom: sample ACF (bars) and $\rho(h) = \phi^h$ (dots)

TSA_ch2_ar_models

Interpreting the AR(1) paths

- $\phi = 0.9$: long swings away from 0; sample $\hat{\rho}(1) = 0.88$, variance 4.98 (theory 5.26)
  - strong persistence looks like a trend over short windows
- $\phi = 0.3$: close to white noise; only $\hat{\rho}(1) = 0.35$ is clearly outside the band
  - variance 1.18 (theory 1.10)
- $\phi = -0.7$: zig-zag; $\hat{\rho}(1) = -0.58$, $\hat{\rho}(2) = 0.34$, alternating signs
  - in economics: overshooting and correction, or a measurement error that is corrected next period
- With $T = 200$ the sample ACF is below the theoretical one at long lags: a small-sample bias that also affects the estimators (Section 5)

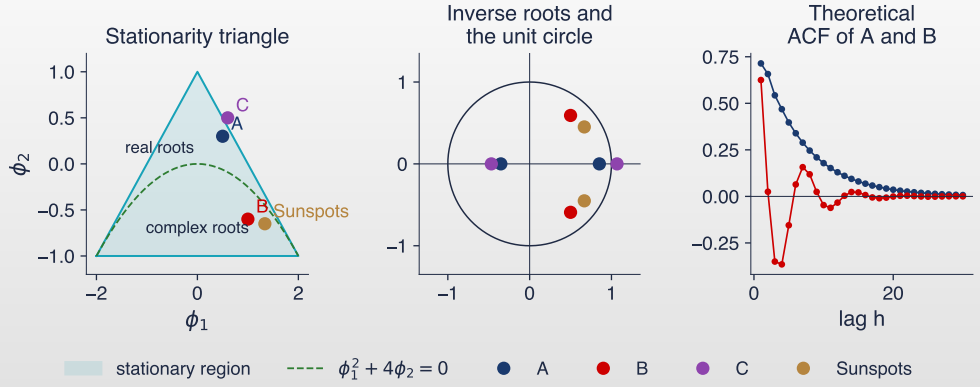
The AR(p) model and its characteristic polynomial

- $\phi(L)X_t = c + \varepsilon_t$, with $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$, a polynomial in a complex number z
  - factorise: $\phi(z) = (1 - \lambda_1 z)\cdots(1 - \lambda_p z)$, where $z_i = 1/\lambda_i$ are the roots
- **Stationarity (causality) condition**: all roots of $\phi(z) = 0$ lie **outside the unit circle**, $|z_i| > 1$
  - equivalently: all **inverse roots** λ_i lie inside, $|\lambda_i| < 1$ (the form shown by software)
  - reason: $1/(1 - \lambda L) = \sum_j \lambda^j L^j$ converges only if $|\lambda| < 1$; each factor is an AR(1)
- A root on the circle ($|z| = 1$, e.g. $z = 1$) is a **unit root**: non-stationarity, the topic of Chapter 3
  - quick necessary check: $\phi_1 + \cdots + \phi_p < 1$ (i.e. $\phi(1) > 0$)

AR(2): the stationarity triangle

- $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ is stationary if and only if
  - $\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad |\phi_2| < 1$
- The roots are complex when $\phi_1^2 + 4\phi_2 < 0$
  - then the ACF is a **damped cosine**: pseudo-cycles of average length $2\pi / \arccos\left(\phi_1 / (2\sqrt{-\phi_2})\right)$
  - damping factor per period: $\sqrt{-\phi_2}$, the modulus of the inverse roots
- **Yule–Walker equations** for the ACF: multiply by X_{t-h}, take expectations, divide by $\gamma(0)$
  - $\rho(1) = \phi_1 + \phi_2 \rho(1)$, so $\rho(1) = \phi_1 / (1 - \phi_2)$
  - $\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2)$ for $h \geq 2$: the ACF obeys the same recursion as the process

Three AR(2) processes and the sunspots



- A: $\phi = (0.5, 0.3)$; B: $(1.0, -0.6)$; C: $(0.6, 0.5)$; Sunspots: Yule's AR(2), estimated in Section 10
- Middle: inverse roots λ_i; stationary means inside the circle

TSA_ch2_ar_models

Interpreting the AR(2) examples

- **A**: real roots -2.84 and 1.17, both outside the circle: stationary
  - $\rho(1) = 0.5/0.7 = 0.714$, $\rho(2) = 0.5\rho(1) + 0.3 = 0.657$: slow, smooth decay
- **B**: complex roots $0.833 \pm 0.986\,i$, modulus $1.291 > 1$: stationary
  - ACF: a wave with period 7.2 lags, damped by 0.775 per lag: a **stochastic cycle**
- **C**: $\phi_1 + \phi_2 = 1.1 > 1$; one root $0.936 < 1$: **not** stationary (explosive)
  - each coefficient is below 1, yet the process explodes: always check the roots, not the coefficients one by one
- Complex roots are how linear models produce business cycles and the 11-year solar cycle

The PACF of an AR(p) cuts off after lag p

- ϕ_{hh} = the last coefficient in the best linear prediction of X_t from $X_{t-1}, \ldots, X_{t-h}$ (Chapter 1)
  - computed from the ACF by the Durbin–Levinson recursion (Durbin (1960))
- For an AR(p) and $h > p$, the best predictor is the model itself: $\phi_1 X_{t-1} + \cdots + \phi_p X_{t-p}$
  - so $\phi_{hh} = 0$ for $h > p$, and $\phi_{pp} = \phi_p$
- **Identification rule**: ACF decays gradually, PACF has p spikes and then stays inside the band: AR(p)
  - band for the sample PACF of an AR(p) at lags $h > p$: $\pm 1.96/\sqrt{T}$

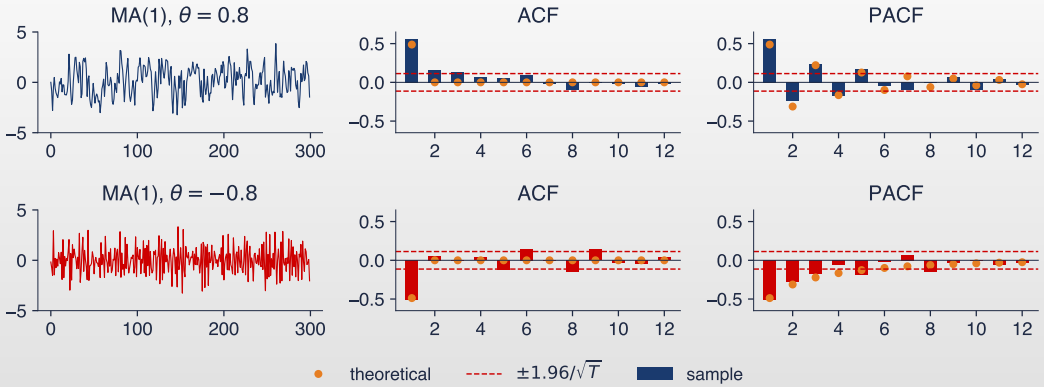
Recap: Autoregressive models

- AR(p): $\phi(L)X_t = c + \varepsilon_t$; stationary iff all roots of $\phi(z)$ lie outside the unit circle
- Mean $\mu = c/\phi(1)$; ACF from the Yule–Walker recursion, a mix of geometric decays and damped waves
- AR(1): $\rho(h) = \phi^h$, $\psi_j = \phi^j$, half-life $\ln 0.5 / \ln \phi$
- PACF cuts off after lag p: the signature of an AR(p)

The MA(q) model

- $X_t = \mu + \theta(L)\varepsilon_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q}$
  - a finite Wold sum: **always stationary**, for any θ
- Moments (derived in Seminar 1 and Chapter 1): with $\theta_0 = 1$
  - $E[X_t] = \mu$; $\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j\theta_{j+h}$ for $0 \le h \le q$; $\gamma(h) = 0$ for $h > q$
- **Identification rule**: the ACF **cuts off** after lag q; the PACF decays gradually
  - the mirror image of the AR(p) rule
  - band for the sample ACF at lags $h > q$ (Bartlett): $\pm 1.96\sqrt{(1 + 2\sum_{j=1}^{q} \hat{\rho}(j)^2)/T}$, wider than $\pm 1.96/\sqrt{T}$

MA(1): $\theta = 0.8$ and $\theta = -0.8$



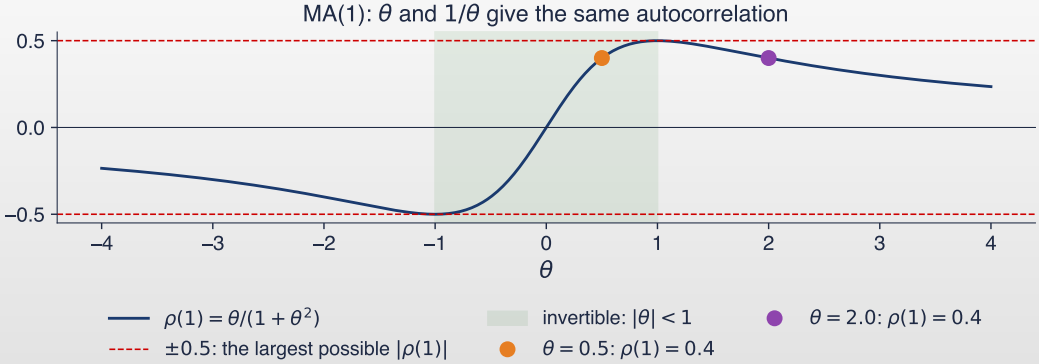
- $T = 300$; the ACF has one spike, the PACF decays (with alternating signs when $\theta > 0$)

TSA_ch2_ma_models

Interpreting the MA(1) correlograms

- Theory: $\rho(1) = \theta/(1+\theta^2) = \pm 0.488$, $\rho(h) = 0$ for $h \geq 2$
  - sample: $\hat{\rho}(1) = 0.56$ and -0.51
- The PACF does not cut off: $\hat{\phi}_{22} = -0.23$ for $\theta = 0.8$
  - an MA(1) is an AR(∞) (next slides), so every past value helps a little
- An MA(1) remembers exactly one period: a shock today affects tomorrow and is then forgotten
- Economic examples: overlapping observations (Section 9), a measurement error, a bid–ask bounce in prices

The largest autocorrelation of an MA(1)



⊡ $\rho(1) = \theta/(1+\theta^2)$ as a function of θ

TSA_ch2_ma_models

Interpreting the curve $\rho(1)$

- Maximum $|\rho(1)| = 0.5$ at $\theta = \pm 1$: an MA(1) can never produce $|\rho(1)| > 0.5$
  - if $\hat{\rho}(1) = 0.7$ and then nothing: not an MA(1); try an AR or an MA(q) with more terms
- $\theta = 0.5$ and $\theta = 2$ give the same $\rho(1) = 0.4$
  - in general θ and $1/\theta$ give the same ACF (with σ^2 rescaled by θ^2)
  - the data cannot tell them apart: an **identification problem**
- The convention that removes the ambiguity: invertibility, $|\theta| < 1$ (next slide)

Invertibility

- MA(1): $\varepsilon_t = X_t - \theta\varepsilon_{t-1}$; substitute repeatedly ($\mu = 0$):
  - $\varepsilon_t = X_t - \theta X_{t-1} + \theta^2 X_{t-2} - \cdots = \sum_{j\geq 0}(-\theta)^j X_{t-j}$, if $|\theta| < 1$
  - an **AR(∞) representation**: $X_t = \theta X_{t-1} - \theta^2 X_{t-2} + \cdots + \varepsilon_t$
- **Invertible**: the shocks can be recovered from the past of X; condition: all roots of $\theta(z) = 0$ outside the unit circle
  - MA(1): $|\theta| < 1$; the same algebra as AR stationarity, applied to $\theta(z)$
- Why we want it
  - uniqueness: among θ and $1/\theta$ only one is invertible
  - forecasting: the shocks ε_t are computed from the data, so forecasts can use them
  - software reports invertible estimates; a root near the circle signals over-differencing (Chapter 1)

Worked example: $\theta = 2$ or $\theta = 0.5$?

- $X_t = \varepsilon_t + 2\varepsilon_{t-1}$, $\sigma^2 = 1$; and $Y_t = u_t + 0.5u_{t-1}$, $\mathrm{Var}(u_t) = 4$
- X: $\gamma(0) = 1 \cdot (1 + 4) = 5$, $\gamma(1) = 2 \cdot 1 = 2$
  - Y: $\gamma(0) = 4 \cdot (1 + 0.25) = 5$, $\gamma(1) = 0.5 \cdot 4 = 2$
- Same mean, same autocovariances: with Gaussian shocks, the **same distribution**
  - no amount of data separates them
- Only Y is invertible: $u_t = Y_t - 0.5Y_{t-1} + 0.25Y_{t-2} - \ldots$
  - for X, the same sum with weights $(-2)^j$ explodes
- Rule: report the invertible version ($\theta = 0.5$, $\sigma^2 = 4$)

Recap: Moving-average models

- MA(q) is always stationary; its ACF cuts off after lag q, its PACF decays
- MA(1): $\rho(1) = \theta/(1+\theta^2)$, at most 0.5 in absolute value
- Invertible iff the roots of $\theta(z)$ lie outside the unit circle; then MA = AR(∞)
- θ and $1/\theta$ are observationally equivalent; we keep the invertible one

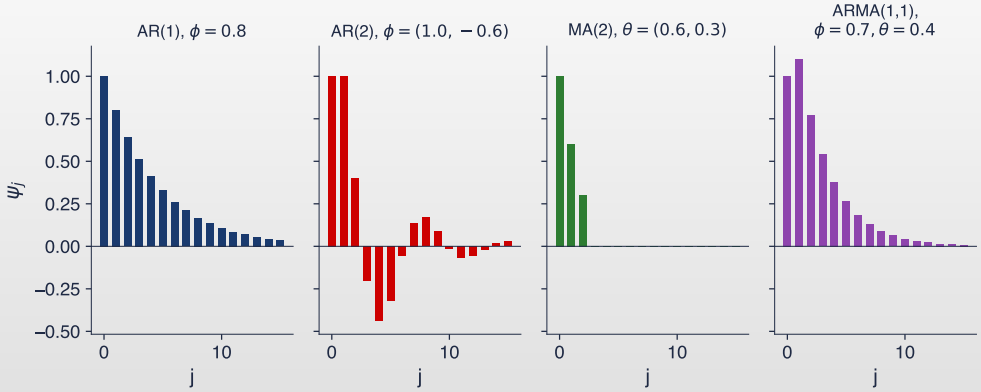
The ARMA(p, q) model

- $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$, i.e.
  - $X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$, $c = \mu\,\phi(1)$
- Three conditions (Brockwell and Davis (2016), Ch. 3)
  - **stationary and causal**: roots of $\phi(z)$ outside the unit circle; then $X_t - \mu = \psi(L)\varepsilon_t$, $\psi(L) = \theta(L)/\phi(L)$
  - **invertible**: roots of $\theta(z)$ outside the unit circle; then $\pi(L)(X_t - \mu) = \varepsilon_t$, $\pi(L) = \phi(L)/\theta(L)$
  - **no common factors**: $\phi(z)$ and $\theta(z)$ share no root
- ARMA(1,1) can match an ACF that decays at rate ϕ from a first value set by θ (below ϕ if $\theta < 0$, above if $\theta > 0$); a pure AR or MA would need many terms for it

The ψ weights and the impulse response

- From $\phi(L)\psi(L) = \theta(L)$, match the coefficients of L^j:
  - $\psi_0 = 1, \quad \psi_j = \theta_j + \sum_{i=1}^{\min(j,p)} \phi_i \psi_{j-i} \quad (\theta_j = 0 \text{ for } j > q)$
- ARMA(1,1): $\psi_1 = \phi + \theta$, $\psi_j = \phi^{j-1}(\phi + \theta)$ for $j \geq 1$
  - example $\phi = 0.7$, $\theta = 0.4$: $\psi_1 = 1.10$, $\psi_2 = 0.77$
- **Impulse response**: $\psi_j = \partial X_{t+j} / \partial \varepsilon_t$, the effect of a unit shock after j periods
  - the same object drives the forecast errors (Section 7) and the impulse responses of VAR models (Chapter 6)

Impulse responses of four models



- ψ_j, $j = 0, \ldots, 15$, from the recursion above

TSA_ch2_arma_acf

Interpreting the impulse responses

- AR(1): geometric decay 0.8^j; the shock never vanishes completely, but fades fast enough for $\sum \psi_j^2 < \infty$
- AR(2) with complex roots: the response changes sign, $\psi_3 = -0.20$, $\psi_4 = -0.44$: a shock creates a cycle
- MA(2): exactly zero after two periods: finite memory
- ARMA(1,1): the MA term lifts the first response to $\psi_1 = 1.10 > 1$; afterwards the AR part takes over

Moments of the ARMA(1,1)

- $X_t = \phi X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$, $|\phi| < 1$, $|\theta| < 1$, $\phi + \theta \neq 0$
- Variance, from $\gamma(0) = \sigma^2 \sum_j \psi_j^2$:
  - $\gamma(0) = \sigma^2 \dfrac{1 + 2\phi\theta + \theta^2}{1 - \phi^2}$; for $\phi = 0.7$, $\theta = 0.4$, $\sigma^2 = 1$: 3.37
- Autocorrelations:
  - $\rho(1) = \dfrac{(1 + \phi\theta)(\phi + \theta)}{1 + 2\phi\theta + \theta^2}$, $\quad \rho(h) = \phi\, \rho(h-1)$ for $h \geq 2$
  - example: $\rho(1) = 0.819$, $\rho(2) = 0.573$
- The ACF decays like an AR(1) from lag 1 onwards, but $\rho(1) \neq \phi$: the MA part only moves the starting point

Common factors: a model that is too large

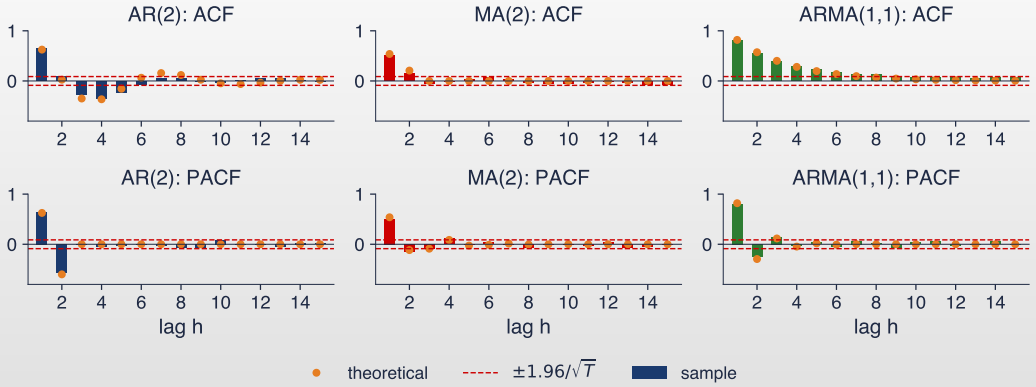
- $X_t = 0.5X_{t-1} + \varepsilon_t - 0.5\varepsilon_{t-1}$
  - $(1 - 0.5L)X_t = (1 - 0.5L)\varepsilon_t$: cancel the factor, $X_t = \varepsilon_t$, white noise
- In general $\phi = -\theta$ in an ARMA(1,1) gives white noise, for any value
  - the likelihood is flat along the line $\phi = -\theta$: the parameters are not identified
- Symptoms in software output
  - large and opposite estimates, e.g. $\hat{\phi} = 0.9$, $\hat{\theta} = -0.88$, with huge standard errors
  - a log-likelihood no better than the smaller model
- Remedy: reduce p and q together; prefer the smallest model with white-noise residuals

Identification: the ACF–PACF table

Model	ACF	PACF
White noise	no significant lag	no significant lag
AR(p)	decays (geometric or damped wave)	**cuts off** after lag p
MA(q)	**cuts off** after lag q	decays
ARMA(p, q)	decays after lag $q - p$	decays after lag $p - q$
Unit root (Ch. 3)	very slow, almost linear decay	$\hat{\phi}_{11} \approx 1$

- The table proposes **candidates**; the information criteria and the diagnostics decide (Sections 5–6)
- Mixed ARMA models are hard to spot by eye: both functions just decay

Theoretical and sample ACF and PACF



- AR(2) $\phi = (1.0, -0.6)$, MA(2) $\theta = (0.6, 0.3)$, ARMA(1,1) $\phi = 0.7$, $\theta = 0.4$; one simulated path of $T = 500$ each

TSA_ch2_arma_acf

Interpreting the three patterns

- AR(2): the PACF has two spikes, $\phi_{22} = \phi_2 = -0.60$, then zero; the ACF is a damped wave
  - identification is easy: read p from the PACF
- MA(2): $\rho(1) = 0.538$, $\rho(2) = 0.207$, then zero; the PACF decays slowly
  - the sample ACF at lag 2 is only just outside the band: with real data, MA(1) and MA(2) can look alike
- ARMA(1,1): both decay; the ACF looks like an AR(1), the PACF has a second, smaller spike
  - an analyst who reads only the PACF would choose AR(2) or AR(3): close, but more parameters
- Sample correlograms are noisy even with $T = 500$; with 100 quarterly observations, much more so

Recap: ARMA models

- ARMA(p, q): $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$; causal if $\phi(z)$ has roots outside the circle, invertible if $\theta(z)$ has
- ψ weights by the recursion $\psi_j = \theta_j + \sum \phi_i \psi_{j-i}$: the impulse response
- No common factors; a flat likelihood and opposite estimates signal one
- ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA

Three estimators

- **Yule–Walker** (method of moments, Yule (1927); Walker (1931)): replace $\rho(h)$ by $\hat{\rho}(h)$ in the Yule–Walker equations
  - pure AR only; fast; always gives a stationary model; less precise when roots are near the circle
- **Conditional least squares**: minimise $\sum_t \varepsilon_t^2$, treating the first values as fixed
  - AR(p): ordinary least squares (OLS) of X_t on $1, X_{t-1}, \ldots, X_{t-p}$
  - MA and ARMA: the residuals $\varepsilon_t = X_t - \mu - \theta\varepsilon_{t-1} - \ldots$ are computed recursively from $\varepsilon_0 = 0$; numerical minimisation
- **Exact Gaussian maximum likelihood** (MLE): maximise the joint density of $(X_1, \ldots, X_T)$
  - uses the first observations too; the default of `statsmodels` `ARIMA` (Kalman filter, Chapter 10)
- In large samples all three agree for pure AR models; MLE is the reference for MA and ARMA

Yule–Walker for an AR(p)

- For $h = 1, \dots, p$: $\rho(h) = \phi_1 \rho(h-1) + \cdots + \phi_p \rho(h-p)$
  - in matrix form $R\,\phi = \rho$, $R_{ij} = \rho(|i-j|)$; $\phi = (\phi_1, \dots, \phi_p)^\top$, $\rho = (\rho(1), \dots, \rho(p))^\top$
  - R: the $p \times p$ matrix of autocorrelations; the hat marks the sample version
  - estimator $\hat{\phi} = \hat{R}^{-1}\hat{\rho}$; noise variance $\hat{\sigma}^2 = \hat{\gamma}(0)(1 - \hat{\phi}^\top \hat{\rho})$
- AR(2) by Cramer's rule:
  - $\hat{\phi}_1 = \dfrac{\hat{\rho}_1(1-\hat{\rho}_2)}{1-\hat{\rho}_1^2}, \quad \hat{\phi}_2 = \dfrac{\hat{\rho}_2 - \hat{\rho}_1^2}{1-\hat{\rho}_1^2}$
  - $\hat{\phi}_2$ is the sample PACF at lag 2 (Chapter 1)



Gilbert Walker (1868–1958), who extended Yule's method in 1931

Photo: unknown author (1925); public domain; Wikimedia Commons

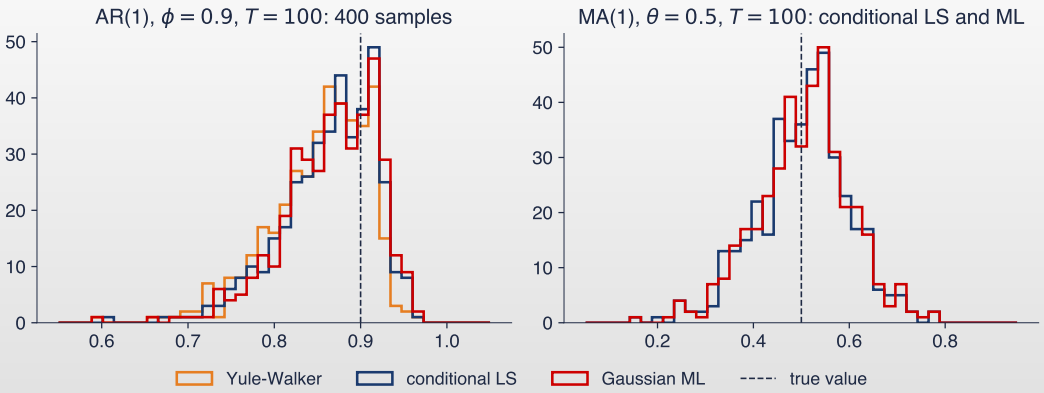
Worked example: Yule–Walker on Romanian GDP growth

- Annual growth of real GDP, 2000Q1–2026Q2 (Section 9): $\hat{\rho}(1) = 0.62$, $\hat{\rho}(2) = 0.40$, $\hat{\gamma}(0) = 18.77$
- Determinant $1 - \hat{\rho}_1^2 = 0.620$
  - $\hat{\phi}_1 = 0.62\,(1 - 0.40)/0.620 = 0.600$
  - $\hat{\phi}_2 = (0.40 - 0.62^2)/0.620 = 0.027$
- $\hat{\sigma}^2 = 18.77\,(1 - \hat{\phi}_1\hat{\rho}_1 - \hat{\phi}_2\hat{\rho}_2) = 11.62$
- $\hat{\phi}_2 \approx 0$: the second lag adds nothing, AR(1) would do
  - but is an AR model the right family at all? Section 9 shows that an MA(3) fits better

Maximum likelihood

- Gaussian likelihood by the **prediction-error decomposition**
  - $f(x_1, \dots, x_T) = \prod_t f(x_t \mid x_{t-1}, \dots, x_1)$; each factor is $N(\hat{x}_{t|t-1}, v_t)$
  - $\ln L = -\frac{1}{2} \sum_{t=1}^{T} \left[\ln(2\pi v_t) + (x_t - \hat{x}_{t|t-1})^2 / v_t \right]$
  - $\hat{x}_{t|t-1}$: one-step prediction; v_t: its error variance, larger for the first observations
- AR(1) by hand: $X_1 \sim N(\mu, \sigma^2/(1-\phi^2))$, then $X_t \mid X_{t-1} \sim N(c + \phi X_{t-1}, \sigma^2)$
  - dropping the first factor gives conditional least squares
- Standard errors (se) from the curvature of $\ln L$; asymptotically
  - AR(1): $\mathrm{se}(\hat{\phi}) \approx \sqrt{(1-\phi^2)/T}$; MA(1): $\mathrm{se}(\hat{\theta}) \approx \sqrt{(1-\theta^2)/T}$
  - t-ratios $\hat{\phi}/\mathrm{se}$ compared with ± 1.96, as in regression
  - `statsmodels` reports by default the **OPG** se (outer product of the scores): the same limit if the model is right, but smaller than the curvature se when the errors have heavy tails; the se in this chapter are OPG

Sampling distributions of the estimators



⊡ 400 simulated samples of $T = 100$ each; left: AR(1) with $\phi = 0.9$; right: MA(1) with $\theta = 0.5$

TSA_ch2_estimation

Interpreting the estimator comparison

- AR(1), $\phi = 0.9$: means 0.853 (Yule–Walker), 0.864 (conditional least squares), 0.866 (MLE)
  - all three are biased downwards; the textbook approximation of the least squares bias is $-(1 + 3\phi)/T = -0.037$
  - Yule–Walker is the most biased near the unit circle
- Standard deviations 0.056, 0.055, 0.054, above the asymptotic 0.044
  - the distribution is skewed to the left: asymptotic intervals are optimistic near $\phi = 1$
- MA(1), $\theta = 0.5$: conditional least squares 0.504 (sd 0.097), MLE 0.506 (sd 0.098); asymptotic sd 0.087
- Practical rule: use MLE; Yule–Walker only for quick starting values or for teaching

Recap: Estimation

- Yule–Walker: moments, pure AR, $\hat{\phi} = \hat{R}^{-1}\hat{\rho}$
- Conditional least squares: OLS for AR, a recursion for MA
- Exact Gaussian MLE: the prediction-error decomposition; the standard in software
- Persistent AR coefficients are biased downwards in small samples

Information criteria

- Fit improves with every parameter; the criteria charge a **penalty** per parameter
  - k = number of estimated parameters (ϕ, θ, μ, σ^2), $\ln L$ = maximised log-likelihood
- **AIC** $= -2 \ln L + 2k$ (Akaike (1974))
  - **AICc** $= \text{AIC} + 2k(k+1)/(T-k-1)$, for small T (Hurvich and Tsai (1989))
- **BIC** $= -2 \ln L + k \ln T$ (Schwarz (1978))
  - $\ln T > 2$ once $T \geq 8$: BIC prefers smaller models
- Choose the model with the **smallest** value; compare only models fitted to the same observations



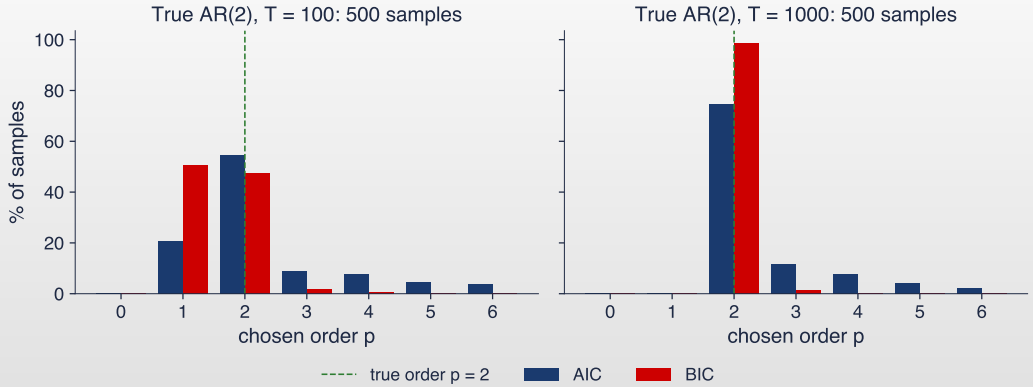
Hirotugu Akaike (1927–2009), Institute of Statistical Mathematics, Tokyo

Photo: The Institute of Statistical Mathematics (2017); CC BY-SA 4.0; Wikimedia Commons

AIC or BIC?

- **BIC is consistent**: if the true model is among the candidates, BIC finds it with probability $\to 1$
  - but it may choose a model that is too small when T is small
- **AIC targets forecasting**: it estimates the expected out-of-sample fit (Kullback–Leibler distance)
  - it overfits with positive probability, even with large T
- Practice (Hyndman and Athanasopoulos (2021), Ch. 9; Hyndman and Khandakar (2008))
  - automatic searches (`auto.arima`, `pmdarima`) minimise AICc over a grid of (p, q)
  - when AIC and BIC disagree, keep both candidates and let the diagnostics and the forecasts decide

How often AIC and BIC find the true order



- True model AR(2), $\phi = (0.5, 0.25)$; AR(p), $p = 0, \ldots, 6$, fitted by OLS on the same sample; 500 samples per panel

TSA_ch2_selection

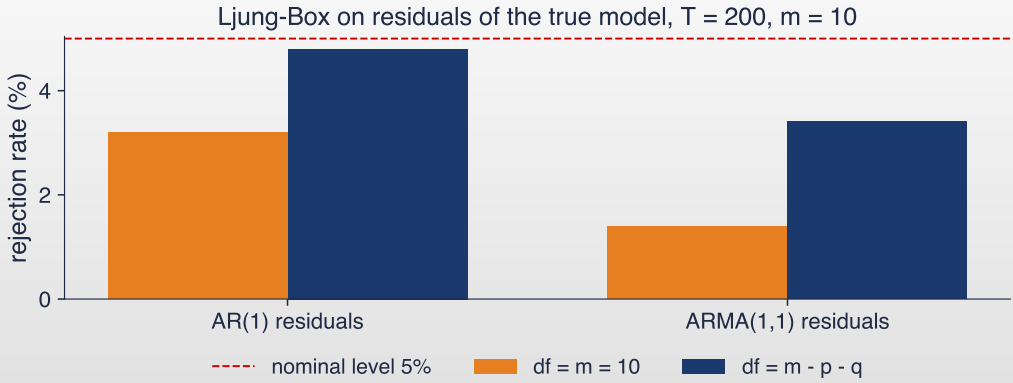
Interpreting the selection experiment

- $T = 100$: AIC finds $p = 2$ in 54.4% of samples, BIC in 47.4%
  - BIC chooses too small a model in 50.4% (the small $\phi_2 = 0.25$ is hard to detect); AIC too large in 24.8%
- $T = 1000$: BIC 98.6%, AIC 74.4%
  - BIC is consistent; AIC still overfits in 25.6% of samples
- Neither criterion is a proof: with quarterly macro data ($T \approx 100$) the order is uncertain
- Overfitting costs little in forecasts; underfitting leaves correlation in the residuals, which the diagnostics catch

Residual diagnostics

- If the model is right, the residuals $\hat{\varepsilon}_t$ behave like white noise
  - plot them: no pattern, no change of variance, no isolated huge values
  - residual ACF inside $\pm 1.96/\sqrt{T}$, except about 1 lag in 20
- **Ljung–Box** on residuals (Ljung and Box (1978)): $Q^*(m) = T(T+2)\sum_{h=1}^{m} \hat{\rho}_{\hat{\varepsilon}}(h)^2/(T-h)$
  - under H_0 (correct model) $Q^*(m) \approx \chi^2(m-p-q)$ (Box and Pierce (1970)): estimating $p+q$ coefficients uses up $p+q$ degrees of freedom
  - choose m well above $p+q$: $m = 10$ for annual or quarterly data, $m = 2s$ for seasonal data (Hyndman and Athanasopoulos (2021))
- **Normality**: Jarque–Bera (Jarque and Bera (1987)) $\text{JB} = \frac{T}{6}\left(S^2 + (K-3)^2/4\right) \approx \chi^2(2)$; QQ plot
  - S = skewness, K = kurtosis (0 and 3 for the Normal distribution); reject normality at 5% if JB > 5.99, the 95% quantile of $\chi^2(2)$
  - non-Normal residuals do not bias $\hat{\phi}$, but they make Normal intervals wrong
- **Squared residuals**: Ljung–Box on $\hat{\varepsilon}_t^2$ detects volatility clustering (Chapter 5)

Why $m - p - q$ degrees of freedom



- Ljung–Box at 5% on the residuals of the **true** model; 1 000 samples of AR(1), half as many of ARMA(1,1)

TSA_ch2_selection

Interpreting the degrees-of-freedom experiment

- With $m = 10$ degrees of freedom the test rejects too rarely: 3.2% (AR(1)) and 1.4% (ARMA(1,1)) instead of 5%
  - residuals are fitted to look uncorrelated at the first lags: their ACF is smaller than that of true white noise
- With $m - p - q$: 4.8% and 3.4%, close to the nominal 5%
  - the correction matters most when $p + q$ is large relative to m
- Software (`acorr_ljungbox`) uses m by default: pass `model_df = p + q`
- Wrong degrees of freedom make a bad model look acceptable

Recap: Model selection and diagnostics

- AIC $= -2 \ln L + 2k$, BIC $= -2 \ln L + k \ln T$; smallest wins; BIC picks smaller models
- BIC is consistent, AIC aims at forecasting and overfits sometimes
- Residuals: plot, ACF, Ljung–Box with $m - p - q$ degrees of freedom, Jarque–Bera, Ljung–Box on squares
- A model passes when its residuals are white noise; normality matters for the intervals

The optimal forecast

- Forecast of X_{T+h} made at time T: $\hat{X}_{T+h|T} = E[X_{T+h} \mid X_T, X_{T-1}, \dots]$
  - it minimises the mean squared error (MSE) among all functions of the past
- Recipe: write the model at $T + h$ and replace
  - future shocks ε_{T+j}, $j \geq 1$, by 0
  - future values X_{T+j} by their forecasts; past values and past residuals stay as they are
- AR(1): $\hat{X}_{T+h|T} = \mu + \phi^h(X_T - \mu)$: **mean reversion** at the speed ϕ
  - MA(q): $\hat{X}_{T+h|T} = \mu$ for $h > q$: after q steps the model knows nothing beyond the mean

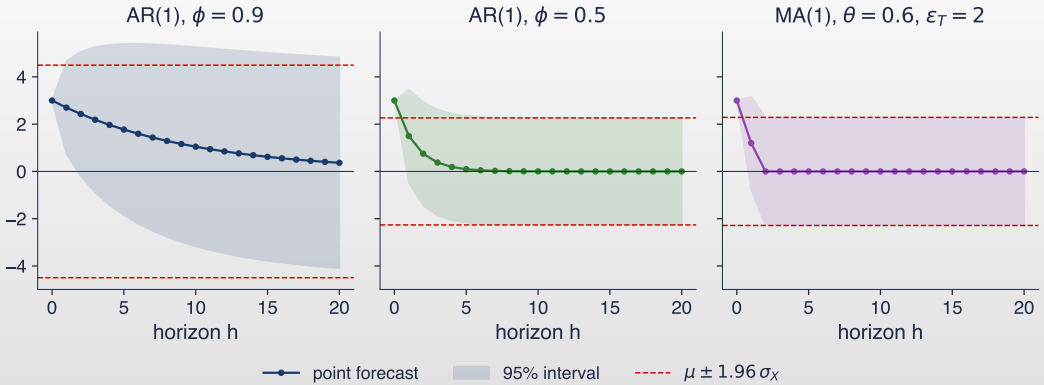
Forecast errors and intervals

- From the causal form $X_{T+h} = \mu + \sum_j \psi_j \varepsilon_{T+h-j}$:
  - error $e_{T+h} = X_{T+h} - \hat{X}_{T+h|T} = \varepsilon_{T+h} + \psi_1 \varepsilon_{T+h-1} + \cdots + \psi_{h-1} \varepsilon_{T+1}$
  - variance $\sigma_h^2 = \sigma^2(1 + \psi_1^2 + \cdots + \psi_{h-1}^2)$, increasing in h
- **95% interval**: $\hat{X}_{T+h|T} \pm 1.96\,\sigma_h$ (Gaussian shocks)
  - $h = 1$: $\pm 1.96\,\sigma$; $h \to \infty$: $\sigma_h^2 \to \gamma(0)$, the unconditional variance
- What the interval leaves out
  - parameter uncertainty ($\hat{\phi}$ instead of ϕ), model uncertainty, fat tails: real coverage is usually below 95%
- Errors of forecasts made at the same time for horizons h and $h+1$ are correlated: they share shocks

Worked example: forecasting an AR(1)

- $X_t = 2 + 0.8X_{t-1} + \varepsilon_t$, $\sigma^2 = 1$, $\mu = 10$; last value $X_T = 12$
- Point forecasts
  - $\hat{X}_{T+1} = 10 + 0.8 \cdot 2 = 11.60$; $\hat{X}_{T+2} = 10 + 0.64 \cdot 2 = 11.28$; $\hat{X}_{T+10} = 10.21$
- Error variances: $\sigma_1^2 = 1$, $\sigma_2^2 = 1 + 0.8^2 = 1.64$
  - 95% intervals: $h = 1$: $[9.64, 13.56]$; $h = 2$: $[8.77, 13.79]$
- Long run: forecast $\to 10$, standard deviation $\to \sqrt{\gamma(0)} = 1.67$

Mean reversion in forecasts



- Start at $X_T = 3$, $\mu = 0$, $\sigma = 1$; dashed: the unconditional band $\mu \pm 1.96\,\sigma_X$

TSA_ch2_forecasting

Interpreting the forecast paths

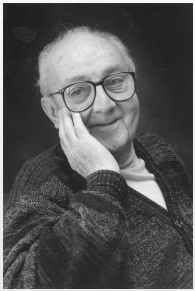
- $\phi = 0.9$: slow reversion; after 5 periods the forecast is still 1.77, standard error 1.85
  - the interval approaches $\pm 1.96 \times 2.29$ only after many periods
- $\phi = 0.5$: after 5 periods the forecast is 0.094, practically the mean
- MA(1): one informative step ($\theta \varepsilon_T = 1.2$), then the mean; the interval jumps to its long-run width at $h = 2$
- A stationary model always forecasts a return to the mean: useful for spreads and growth rates, misleading if the mean shifts

Recap: Forecasting

- $\hat{X}_{T+h|T}$: replace future shocks by 0 and future values by forecasts
- Error variance $\sigma^2 \sum_{j<h} \psi_j^2$; intervals widen up to $\pm 1.96\sqrt{\gamma(0)}$
- Forecasts revert to μ; MA(q) forecasts equal μ beyond q steps
- Intervals ignore parameter and model uncertainty

George Box and Gwilym Jenkins

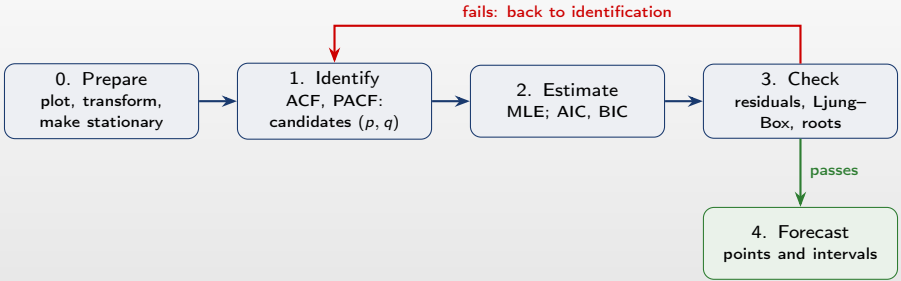
- **George E. P. Box** (1919–2013), statistician, University of Wisconsin–Madison
  - "All models are wrong, but some are useful"
- **Gwilym M. Jenkins** (1932–1982), statistician, Lancaster University
  - *Time Series Analysis: Forecasting and Control*, 1970; fourth edition with Reinsel: Box, Jenkins and Reinsel (2008)
- Their contribution: not new models, but a **method**
  - a loop of identification, estimation and checking that anyone can apply
  - Box–Pierce (1970) and Ljung–Box (1978) come from this programme



George E. P. Box

Photo: DavidMCEddy; CC BY-SA 3.0; Wikimedia Commons

The Box–Jenkins loop



- Step 0 is Chapters 1 and 3: logs, differences, unit-root tests; ARMA needs a stationary series
- The loop ends with the smallest model whose residuals are white noise and whose roots are well inside the admissible region
- Out-of-sample checks (Chapter 0: training and test samples) complete the method

The data

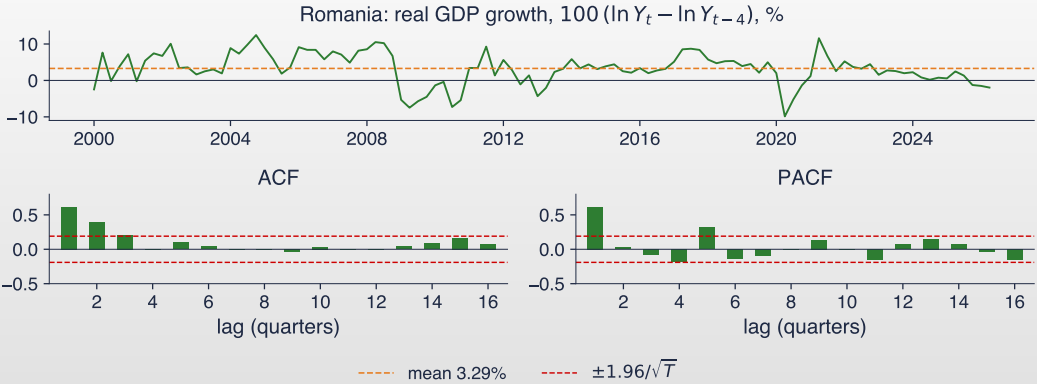
- Romanian real GDP, chain-linked volumes (2010 prices), quarterly, not seasonally adjusted; Eurostat, `namq_10_gdp`
  - the national source: the National Institute of Statistics (INS)
- Annual growth $y_t = 100\,(\ln Y_t - \ln Y_{t-4})$, in %: removes the season (Chapter 1)
  - 2000Q1–2026Q2: $T = 106$ quarters, mean 3.29%, standard deviation 4.35%
  - extremes: -9.9% in 2020Q2, $+12.5\%$ in 2004Q4; last value -2.0%
- Stationary? No trend in the growth rate; a formal test is in Chapter 3



The National Institute of Statistics, Bucharest

Photo: Dan Mihai Pitea (2009); CC BY-SA 3.0; Wikimedia Commons

Step 1: identification



⊡ Annual growth of Romanian real GDP, its ACF and PACF with $\pm 1.96/\sqrt{T} = \pm 0.19$

TSA_ch2_gdp_case

Interpreting the correlograms of GDP growth

- ACF: 0.62, 0.40, 0.21, then -0.01 at lag 4: **cuts off after lag 3**
  - candidate: MA(3)
- PACF: 0.62 at lag 1, then small, except 0.32 at lag 5
  - candidate: AR(1); lag 5 may be chance or a remnant of the season
- Why MA(3)? y_t is the sum of four quarterly growth rates, $y_t \approx g_t + g_{t-1} + g_{t-2} + g_{t-3}$
  - if g_t is close to white noise (Seminar 2, B2), consecutive annual rates share three quarters: an MA(3) by construction
- Ljung–Box $Q^*(8) = 64.9$ on the series itself: strongly autocorrelated, so a model is needed

Step 2: estimation and information criteria

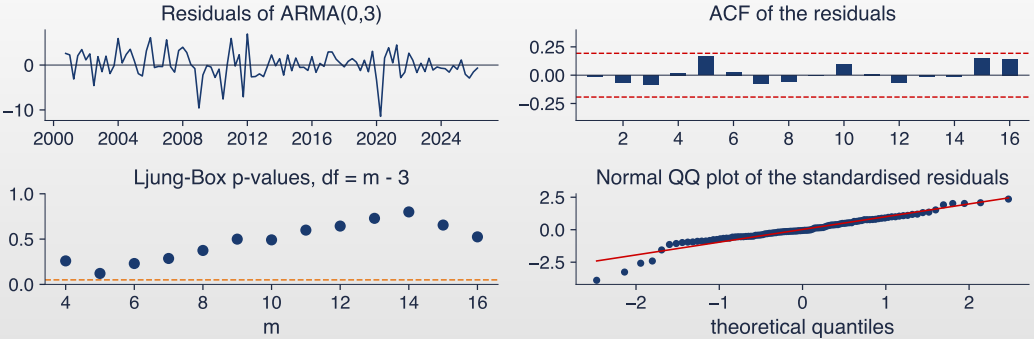
Model	k	$\ln L$	AIC	BIC	$\hat{\sigma}$	LB p-value
ARMA(0,0)	2	−305.83	615.66	620.99	4.33	< 0.001
ARMA(1,0)	3	−279.67	565.33	573.32	3.38	0.023
ARMA(2,0)	4	−279.66	567.32	577.98	3.38	0.037
ARMA(1,1)	4	−279.66	567.32	577.98	3.38	0.013
ARMA(0,3)	5	−270.02	550.04	563.36	3.07	0.375
ARMA(1,3)	6	−268.48	548.97	564.95	3.02	0.648
ARMA(2,3)	7	−267.94	549.89	568.53	3.00	0.543

- Exact Gaussian MLE on the same 106 quarters; LB = Ljung–Box $Q^*(8)$ on the residuals, $8 - p - q$ degrees of freedom; full grid $p, q \leq 3$ in the Quantlet
- **BIC** chooses MA(3); **AIC** chooses ARMA(1,3), lower by only 1.07
- AR(1), the PACF candidate, leaves autocorrelation: LB p-value 0.023

Interpreting the estimated MA(3)

- $\hat{y}_t = 3.17 + \hat{\varepsilon}_t + 0.834\,\hat{\varepsilon}_{t-1} + 0.675\,\hat{\varepsilon}_{t-2} + 0.591\,\hat{\varepsilon}_{t-3}$, $\hat{\sigma} = 3.07$
  - standard errors (OPG): 1.03 (mean), 0.083, 0.102, 0.082: all three MA terms are significant
- The $\hat{\theta}_j$ decline from 0.834 to 0.591: close to, but below, the overlap pattern $(1, 1, 1)$
  - $(1, 1, 1)$ would put roots on the unit circle; the smallest root modulus here is $1.17 > 1$: invertible
- ARMA(1,3): $\hat{\theta}_1 = 1.019 > 1$, yet the roots have modulus at least 1.10: still invertible
  - for $q > 1$, invertibility is read from the roots, not from single coefficients
  - $\hat{\phi} = -0.284$ (se OPG 0.143): one extra parameter for little gain; BIC drops it
- Mean growth 3.17% per year, but with a standard error of 1.03: overlapping data carry less information than T suggests

Step 3: diagnostics of the MA(3)



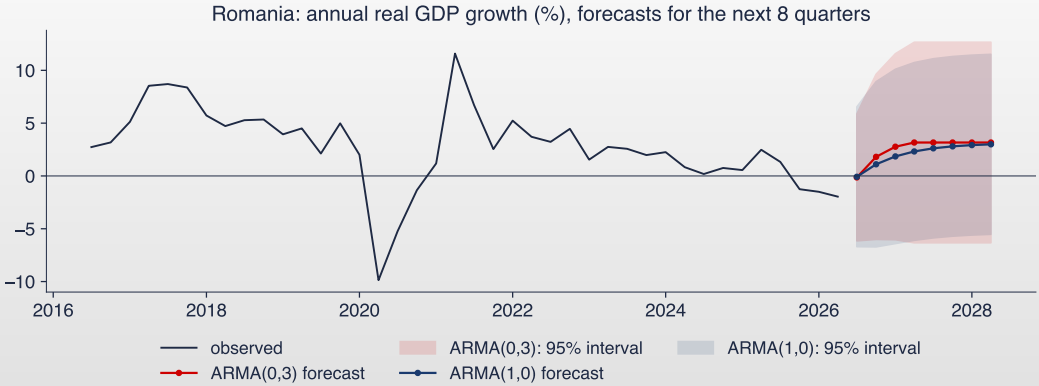
- Residuals, their ACF, Ljung–Box p-values for $m = 4, \ldots, 16$ with $m - 3$ degrees of freedom, Normal QQ plot

TSA_ch2_gdp_case

Interpreting the MA(3) diagnostics

- No residual autocorrelation: $Q^*(8) = 5.35$ on 5 degrees of freedom, $p = 0.375$; $Q^*(16)$: $p = 0.525$
  - every p-value in the chart is above 0.12; with 8 instead of 5 degrees of freedom, p would be 0.720, too reassuring
- Not Normal: Jarque–Bera 39.0, $p < 0.001$; skewness -0.75, kurtosis 5.61
  - the largest residual: -11.4 in 2020Q2, -3.89 standard deviations: the pandemic lockdown
- Squared residuals: Ljung–Box $p = 0.779$, no volatility clustering at this frequency
- Verdict: the MA(3) captures the dependence; the Normal intervals understate the risk of recessions

Step 4: forecasts



- MA(3) (BIC) and AR(1) forecasts for 2026Q3–2028Q2, with 95% intervals

TSA_ch2_gdp_case

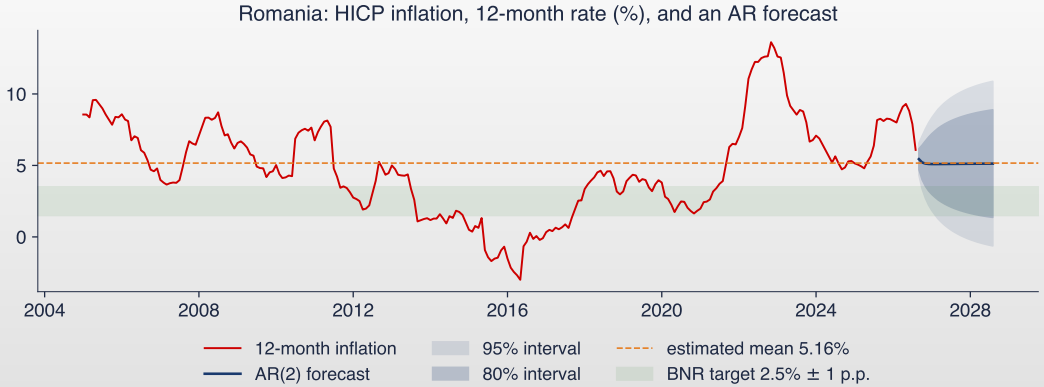
Interpreting the GDP forecasts

- MA(3): -0.14, 1.81, 2.77, then the mean 3.17% from the fourth quarter on
  - exactly the MA(q) rule: after $q = 3$ steps the model only knows the mean
- AR(1): a smooth return to its mean, 2.99% after 8 quarters (mean 3.12%)
  - the two models agree in the short run and differ in how fast they forget the recent slowdown
- Intervals: $[-6.1, 5.9]$ one quarter ahead, $[-6.3, 12.7]$ from four quarters on: ± 6.0 and ± 9.5 percentage points
- A univariate model cannot anticipate a turning point; it tells us how unusual the next value would be

Recap: Romanian GDP growth

- Annual growth of quarterly GDP: ACF cuts off after lag 3, MA(3) by BIC, ARMA(1,3) by AIC
- The MA(3) is the overlap of four quarters: the data structure explains the model
- White-noise residuals but fat tails (2020): intervals are too narrow in crises
- Forecasts reach the mean after three quarters

Romanian inflation: an AR(2) forecast



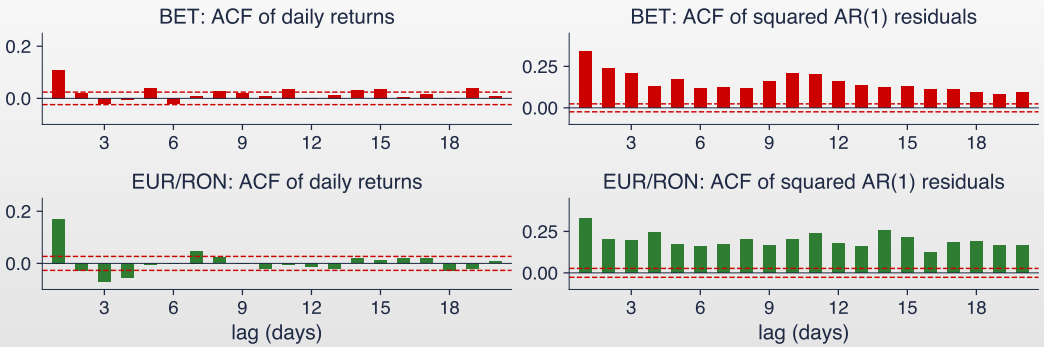
- HICP (harmonised index of consumer prices, Eurostat), 12-month rate in %, 260 months to August 2026; AR order chosen by BIC among $p \leq 6$

TSA_ch2_real_series

Interpreting the inflation model

- BIC: 495.7 for AR(1), 469.7 for AR(2), 473.4 for AR(3): AR(2)
  - $\hat{\phi}_1 = 1.320$ (0.057), $\hat{\phi}_2 = -0.344$ (0.055); $\hat{\phi}_1 + \hat{\phi}_2 = 0.976$, largest inverse root 0.963
- Very close to a unit root: is inflation stationary at all? The test is in Chapter 3
  - the estimated mean, 5.16% (se 1.35), is the average of 2005–2026, above the BNR target of 2.5% $\pm$ 1 p.p.
- Forecast: from 6.09% to 5.44% next month and 5.13% in August 2028; 95% interval $[-0.6, 10.9]$
  - the model reverts to its historical mean, not to the target: it knows nothing about monetary policy
- Diagnostics fail: Ljung–Box $Q^*(12) = 71.6$, p < 0.001: correlation at lag 12 (base effects of the 12-month rate), a job for Chapter 4

Daily returns: BET and EUR/RON



- Left: ACF of daily log returns; right: ACF of the squared residuals of an AR(1)

TSA_ch2_real_series

Interpreting the AR(1) for daily returns

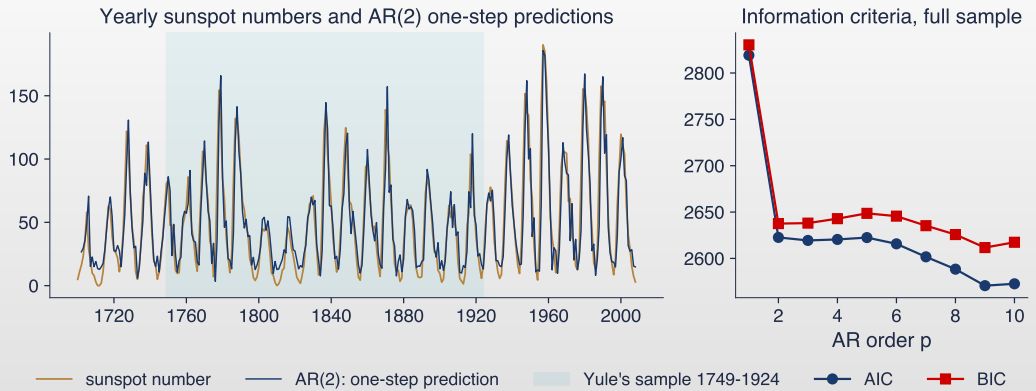
	BET	EUR/RON
Observations	6 670	5 352
$\hat{\phi}$ (se OPG)	0.109 (0.005)	0.170 (0.006)
t-ratio	20.9	30.6
R^2 (%)	1.2	2.9
LB $Q^*(10)$ residuals, 9 df (p)	29.7 (<0.001)	61.5 (<0.001)
LB $Q^*(10)$ squared residuals	2496	2351
Kurtosis of residuals	14.0	17.0
BIC choice, $p, q \leq 2$	ARMA(1,0)	ARMA(2,1)

- se: OPG; with heavy tails (kurtosis 14.0 and 17.0) they are too small, and curvature or robust se are several times larger, yet $\hat{\phi}$ stays significant
- Highly significant $\hat{\phi}$, but R^2 of 1–3%: **statistically** real, **economically** small after trading costs
- Positive $\hat{\phi}$: slow price adjustment (BET), a managed exchange rate (EUR/RON)
- Squared residuals and kurtosis: the mean is modelled, the variance is not; ARMA + GARCH in Chapter 5

Case study: Yule (1927) and the sunspots

- Yule (1927) fitted the first autoregression: Wolfer's yearly sunspot numbers, 1749–1924
  - until then, cycles were modelled as sums of sine waves (periodogram)
  - Yule: a "pendulum bombarded by peas": a damped oscillator kicked by random shocks
- Our least squares AR(2) on the same years (176 observations, statsmodels data):
  - $\hat{\phi}_1 = 1.336$, $\hat{\phi}_2 = -0.650$: complex roots, modulus of the inverse roots 0.806
  - implied period $2\pi / \arccos(\hat{\phi}_1 / (2\sqrt{-\hat{\phi}_2})) = 10.57$ years: the solar cycle, from two coefficients
- Walker (1931) generalised the equations to AR(p): the Yule–Walker equations
  - the start of parametric time series analysis

The sunspots: AR(2) and the information criteria



⊡ Yearly sunspot numbers, 1700–2008 (309 years); AR(p) by MLE on the full sample

TSA_ch2_real_series

Interpreting the sunspot models

- Full-sample AR(2): $\hat{\phi} = (1.391, -0.689)$, close to Yule's; it explains 83% of the variance one year ahead
  - the one-step predictions follow the cycle, but lag behind at the peaks
- AIC and BIC both choose $p = 9$ over $p \leq 10$
  - the cycle is asymmetric (fast rise, slow decay), which a linear AR(2) cannot reproduce
- Lesson: two parameters capture the essence; more lags capture details; nonlinearity is beyond ARMA

Case study: forecasting competitions

- Makridakis and Hibon (2000): the M3 competition, 3003 series (yearly, quarterly, monthly, other), 24 methods
  - forecasts compared out of sample, on several accuracy measures and horizons
- Conclusions confirmed from the earlier M-competitions
  - statistically sophisticated methods do not necessarily forecast better than simple ones
  - the ranking depends on the accuracy measure and on the horizon; combinations of methods do well
- Consequences for ARMA
  - the Box–Jenkins loop must end with an out-of-sample comparison against simple benchmarks (mean, last value, exponential smoothing from Chapter 0)
  - Seminar 2, C1 runs this comparison for Romanian inflation

Recap: More real series

- Inflation: AR(2) near a unit root; reverts to the historical mean; lag-12 residual correlation
- Daily returns: significant but tiny AR effects; the variance needs GARCH
- Sunspots: Yule's AR(2) gives the 11-year cycle from complex roots
- Forecasting competitions: always compare with simple benchmarks

Possible contribution of AI

- **Code**: a first draft of a loop that fits ARMA(p, q) on a grid, tabulates AIC and BIC, and runs the residual tests
- **Explanation**: a second derivation of the ACF of an ARMA(1,1), or of why the PACF of an MA decays
- **Exploration**: the same Box–Jenkins loop on many series (GDP growth of all EU countries) to compare persistence
- Example prompt
  - `Write Python code with statsmodels that fits ARMA(p,q), p,q <= 3, to Romanian annual GDP growth from Eurostat, reports AIC, BIC and the Ljung-Box p-value with model_df = p + q, and forecasts 8 quarters from the BIC model.`

Checks you must run

- The sign convention of θ and ϕ in the software ($\theta(L) = 1 + \theta L$ in statsmodels, $1 - \theta L$ in some books)
- Intercept or mean: the `const` of `ARIMA` is the mean μ, not c
- Ljung–Box degrees of freedom on residuals ($m - p - q$), and that all models are compared on the same sample
- Roots of both polynomials, not the size of single coefficients
- That the series is stationary before an ARMA is fitted (Chapter 3), and that forecasts are compared with simple benchmarks
- Every cited reference: it must exist; check the DOI

Key takeaways

- ARMA(p, q) replaces the infinite Wold sum by a ratio of two short polynomials, $\theta(L)/\phi(L)$
- Stationarity: roots of $\phi(z)$ outside the unit circle; invertibility: roots of $\theta(z)$ outside; no common factors
- ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA
- Estimate by MLE, choose by AIC/BIC, check residuals with Ljung–Box on $m - p - q$ degrees of freedom
- Forecasts revert to the mean; intervals widen to $\pm 1.96\sqrt{\gamma(0)}$ and ignore parameter uncertainty
- Real data: GDP growth is an MA(3) by construction; inflation is near a unit root; returns need GARCH

Key formulas

Quantity	Formula
ARMA(p,q)	$\phi(L)(X_t-\mu)=\theta(L)\varepsilon_t, \quad \psi(L)=\theta(L)/\phi(L)$
AR(1)	$\mu=c/(1-\phi), \quad \gamma(0)=\sigma^2/(1-\phi^2), \quad \rho(h)=\phi^h$
AR(2)	$\rho(1)=\phi_1/(1-\phi_2), \quad \rho(h)=\phi_1\rho(h-1)+\phi_2\rho(h-2)$
MA(1)	$\rho(1)=\theta/(1+\theta^2), \quad \rho(h)=0,\ h\geq 2$; invertible $\lvert\theta\rvert<1$
ARMA(1,1)	$\psi_j=\phi^{j-1}(\phi+\theta), \quad \rho(h)=\phi\rho(h-1),\ h\geq 2$
Yule–Walker	$\hat{\phi}=\hat{R}^{-1}\hat{\rho}, \quad \hat{\sigma}^2=\hat{\gamma}(0)(1-\hat{\phi}^\top\hat{\rho})$
Criteria	$\mathrm{AIC}=-2\ln L+2k, \quad \mathrm{BIC}=-2\ln L+k\ln T$
Residuals	$Q^*(m)\sim\chi^2(m-p-q)$
Forecast	$\hat{X}_{T+h\mid T}\pm 1.96\,\sigma\sqrt{\sum_{j=0}^{h-1}\psi_j^2}$

Self-assessment

- **Question**: is $X_t = 1.2X_{t-1} - 0.32X_{t-2} + \varepsilon_t$ stationary?
  - **Answer**: $1 - 1.2z + 0.32z^2 = (1 - 0.8z)(1 - 0.4z)$, roots 1.25 and 2.5, both outside the circle: yes
- **Question**: the ACF of a series has one spike $\hat{\rho}(1) = -0.45$ and the PACF decays. Which model?
  - **Answer**: MA(1) with $\theta < 0$; solve $\theta/(1 + \theta^2) = -0.45$ and keep the invertible root
- **Question**: an ARMA(2,1) leaves $Q^*(10) = 15.2$ in its residuals. Reject at 5%?
  - **Answer**: degrees of freedom $10 - 3 = 7$, $\chi^2_{0.95}(7) = 14.07 < 15.2$: reject; with 10 degrees of freedom (critical 18.31) one would wrongly accept
- Next: Chapter 3, unit roots and ARIMA models: what to do when a root sits on the unit circle

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